Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 9

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

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1. Work Done

During Week 9, we concentrated on discussing the features of various fabric-defect detection techniques and enhancing our Fourier-defect detection technique. In order to ensure patches from the same image are not split between the data sets, we divided the AITEX patch set according to the images that it originated from. We employed defect-free REF patches to gain an understanding of the normal weave pattern of each fabric, CAL patches to score normalise and threshold calibration, and DEV patches to compare and enhance the methods.

We start by removing the patch mean and linear intensity trend from the main Fourier pipeline. Then we use the Hann window and compute the 2-D FFT. Strong frequency peaks are detected and the repeating weave portions are suppressed using a notch mask. The rest of the spectral energy is then converted to a spatial residual map. This map is smoothed to emphasize parts that have different textures than the common fabric.

Eight different approaches were compared: LBP, LBP self-reference, GLCM, Gabor, FFT fixed single-scale, FFT fixed multi-scale, FFT fixed multi-scale per-fabric, and FFT adaptive multi-scale. The adaptive method is able to determine the main weave period individually for each fabric and choose the windows for analysis.

Then we explored ways to enhance the adaptive method with our DEV split. The notebook explored 63 different sets of window rules, number of scales, smoothing and scale-fusion methods. Various settings for notch-filter, border margin,

contrast equalisation and Gabor fusion were also examined. Once they were compared, c_periods = 8, n_scales = 3, sigma_periods = 1.0, fusion = max, notch_depth = 0.25 and border_margin = 16 pixels were chosen. The refinement objective was reduced for Gabor fusion and hence it wasn't included.

After selecting the final settings, we made the FFT adaptive multi-scale method into a reusable SpectralResidualDetector. It will make use of defect-free REF data to estimate the reference spectra and CAL data to establish the detection thresholds.

We also tested the detector on fabric patches and raw fabric strips. The visual results include original input, deviation map, predicted defect region and the reference mask. Lastly, we combined all the method comparison results saved and ran the frozen detector on TEST while leaving all other settings unchanged.

2. Results

During the tuning process, c_periods = 8, n_scales = 3, sigma_periods = 1.0 and maximum fusion were chosen.

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TUNING RESULTS (63 settings, DEV; defects scored: 163)
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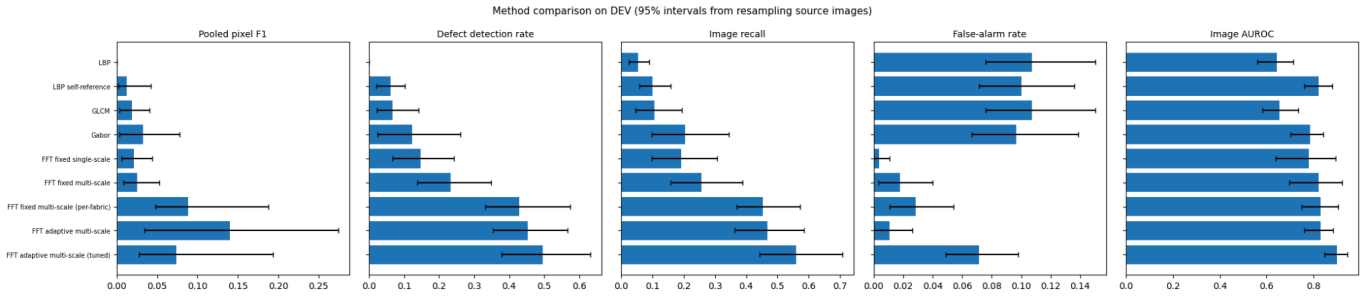
	c_periods	n_scales	sigma_periods	fusion	detect	auc	recall	fpr	img_auc	objective
0	8	3	1.0	max	0.429	0.805	0.507	0.014	0.861	0.617
1	8	3	1.0	rms	0.374	0.809	0.428	0.014	0.874	0.592
2	8	3	0.5	max	0.313	0.806	0.342	0.032	0.807	0.559
3	8	3	1.0	mean	0.294	0.809	0.336	0.011	0.861	0.552
4	4	3	1.0	max	0.374	0.718	0.467	0.011	0.830	0.546
5	2	3	1.0	max	0.374	0.718	0.467	0.011	0.830	0.546
6	8	3	0.5	rms	0.252	0.814	0.296	0.025	0.807	0.533
7	8	3	0.5	mean	0.221	0.816	0.250	0.014	0.799	0.518
8	8	2	1.0	rms	0.288	0.746	0.355	0.011	0.799	0.517
9	4	3	0.5	max	0.313	0.716	0.382	0.018	0.828	0.515

Results in the results table are point estimates with their 95% confidence intervals.

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DEV: 152 defect patches, 280 defect-free patches (95% intervals from 200 resamples of source images)
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metric	Defect detection rate	False-alarm rate	Image AUROC	Image recall	Pooled pixel F1
method					
LBP	0.000 [0.000, 0.000]	0.107 [0.076, 0.150]	0.643 [0.560, 0.714]	0.053 [0.024, 0.090]	0.000 [0.000, 0.000]
LBP self-reference	0.061 [0.020, 0.103]	0.100 [0.071, 0.136]	0.822 [0.761, 0.879]	0.099 [0.058, 0.158]	0.012 [0.002, 0.043]
GLCM	0.067 [0.022, 0.141]	0.107 [0.076, 0.150]	0.656 [0.583, 0.736]	0.105 [0.045, 0.193]	0.019 [0.003, 0.041]
Gabor	0.123 [0.025, 0.262]	0.096 [0.066, 0.139]	0.786 [0.703, 0.840]	0.204 [0.098, 0.344]	0.033 [0.003, 0.078]
FFT fixed single-scale	0.147 [0.067, 0.242]	0.004 [0.000, 0.011]	0.779 [0.638, 0.895]	0.191 [0.097, 0.306]	0.021 [0.005, 0.044]
FFT fixed multi-scale	0.233 [0.139, 0.349]	0.018 [0.003, 0.040]	0.822 [0.697, 0.923]	0.257 [0.159, 0.389]	0.025 [0.008, 0.053]
FFT fixed multi-scale (per-fabric)	0.429 [0.332, 0.574]	0.029 [0.011, 0.054]	0.829 [0.751, 0.907]	0.454 [0.369, 0.573]	0.089 [0.048, 0.188]
FFT adaptive multi-scale	0.454 [0.354, 0.567]	0.011 [0.000, 0.026]	0.830 [0.760, 0.882]	0.467 [0.363, 0.586]	0.141 [0.034, 0.274]
FFT adaptive multi-scale (tuned)	0.497 [0.379, 0.633]	0.071 [0.049, 0.098]	0.900 [0.845, 0.944]	0.559 [0.444, 0.710]	0.074 [0.028, 0.194]

When compared against the other methods, the DEV results indicate that the tuned adaptive method had the highest defect detection rate (0.497), image recall (0.559) and image AUROC (0.900). But it didn't come out the best in all measures. The pooled pixel F1 score for the untuned adaptive method was 0.141, and 0.074 for the tuned adaptive method. The false alarm rate was also higher for the tuned method (0.071 vs. 0.011).This indicates that the tuned method was not the most suitable method for all of the measurements. We chose it for its improvement of a few detection-level metrics and the refinement objective.

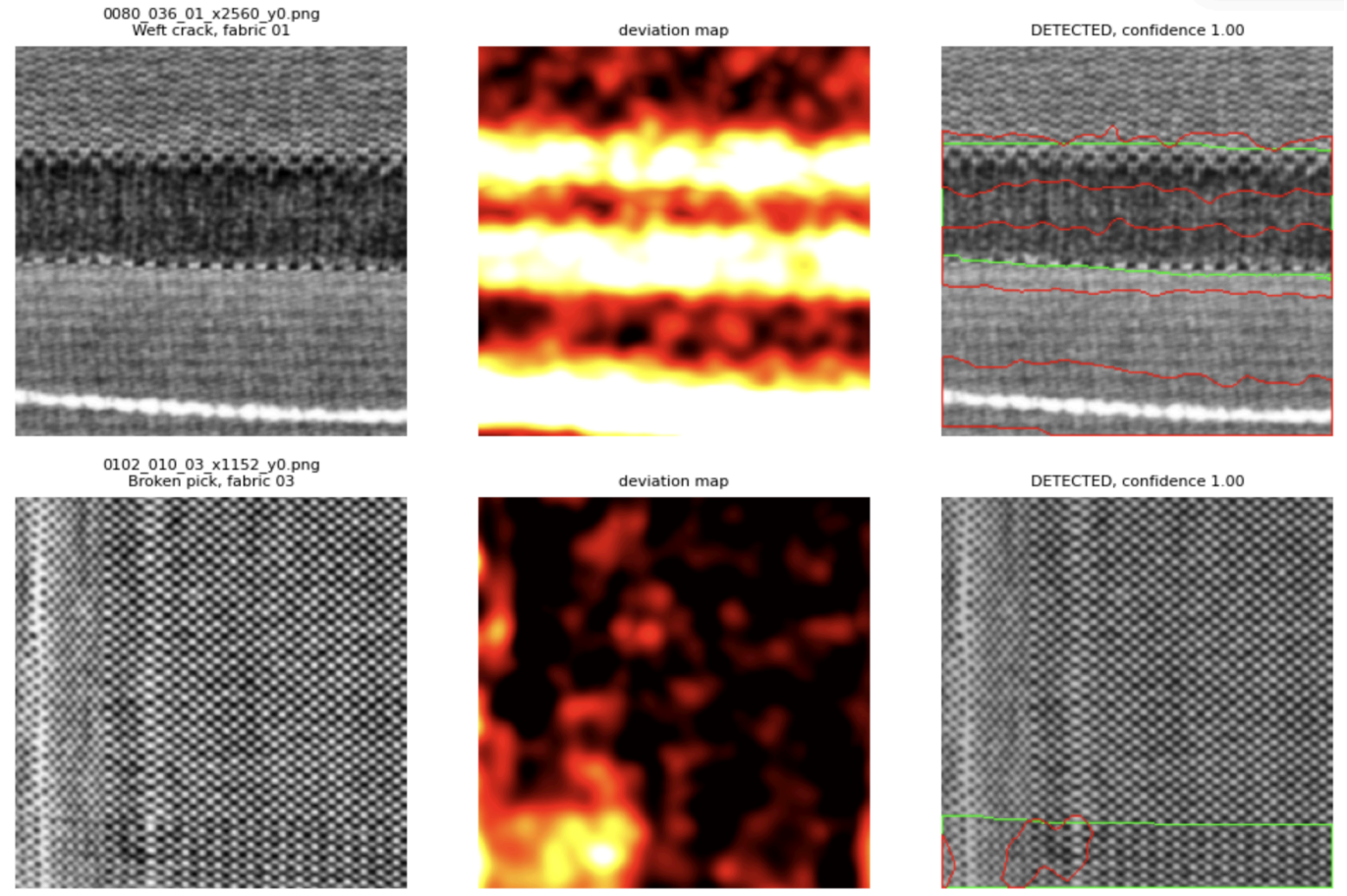


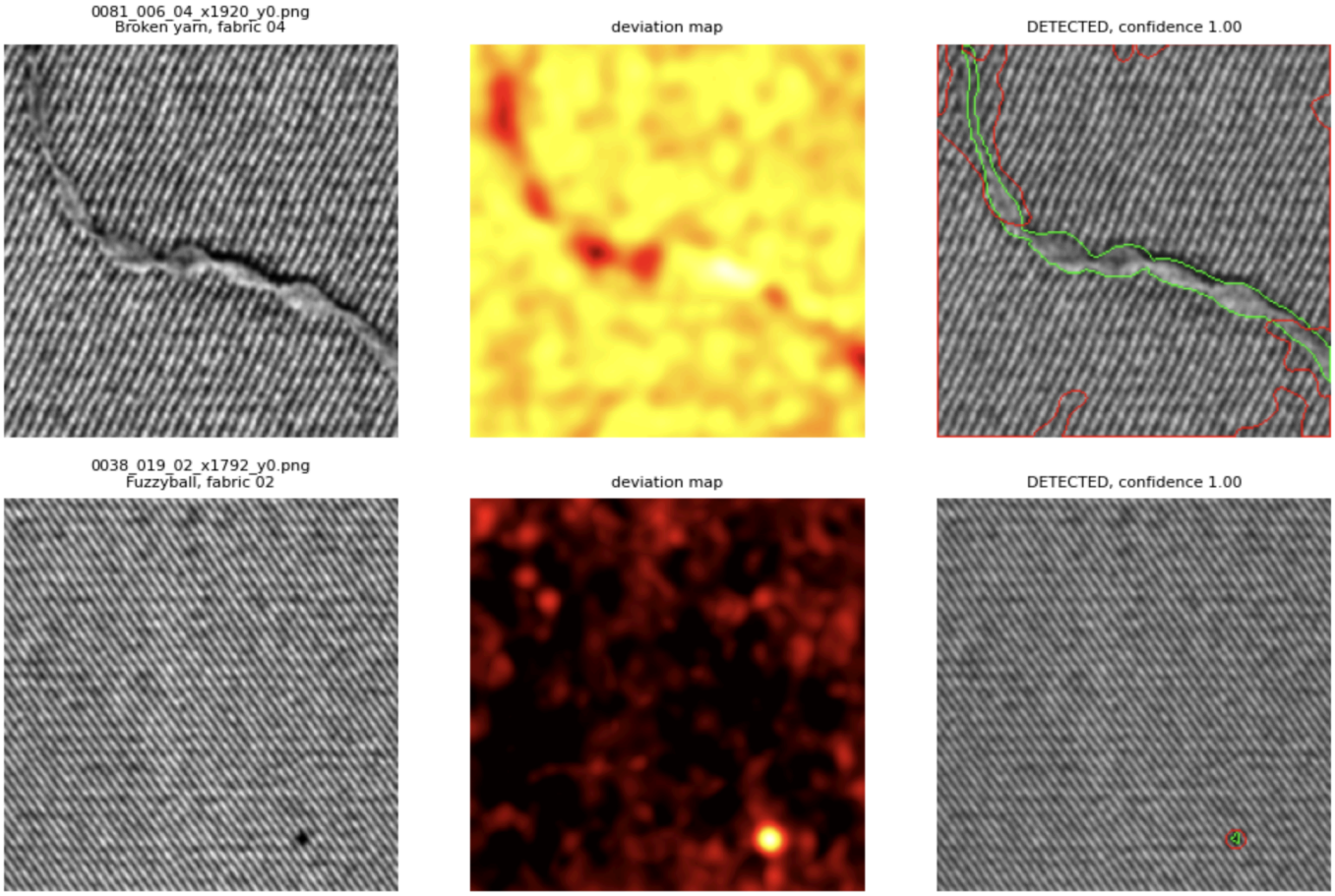
The final detector was also tested in the same way as the saved DEV results. Both gave the same result, the greatest difference being 0.0000.

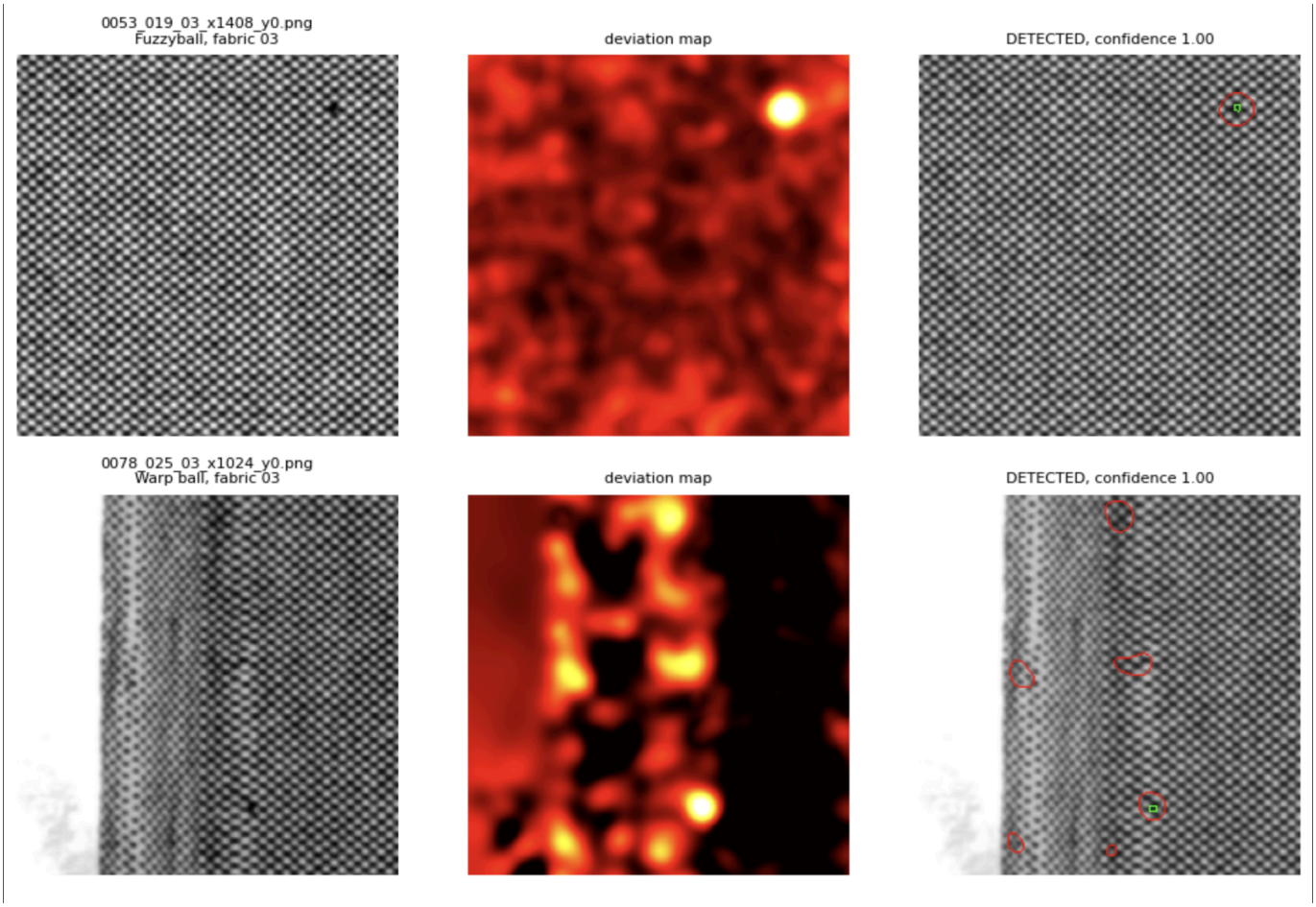
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RESULTS OF THIS RUN, AND COMPARISON WITH THE SAVED EVALUATION
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adaptive:
                        DEV   TEST
Pooled pixel F1       0.074  0.435
Defect detection rate 0.497  0.682
Image recall          0.559  0.678
False-alarm rate      0.071  0.071
Image AUROC           0.900  0.932
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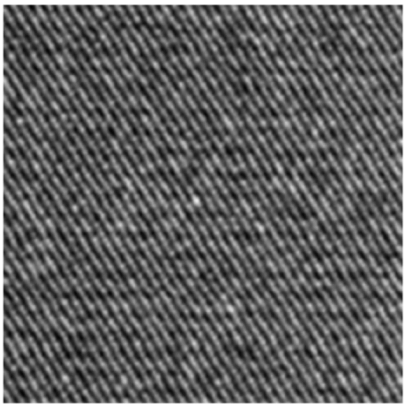
Once the settings are complete, the detector was used on the TEST split. The TEST results were not used for selecting and/or adjusting parameters. This enabled us to test the final method with data not used for the refinement. It is therefore the tuned FFT adaptive multi-scale method (see above) that is selected.

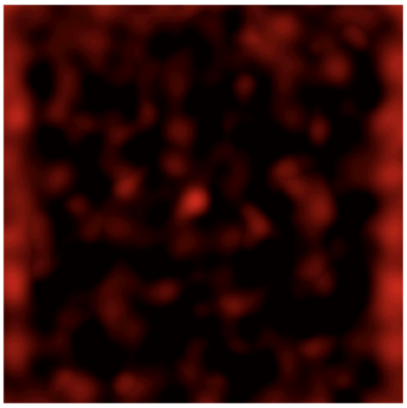



0081_006_04_x1920_y0.png
Broken yarn, fabric 04
deviation map
DETECTED, confidence 1.00
0038_019_02_x1792_y0.png
Fuzzyball, fabric 02
deviation map
DETECTED, confidence 1.00

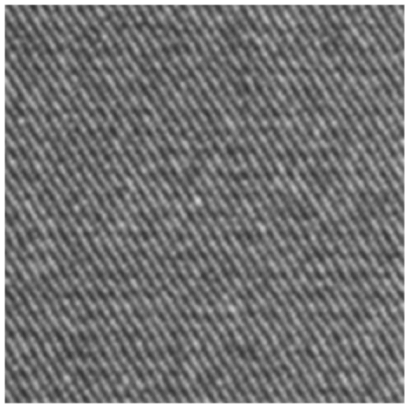

0053_019_03_x1408_y0.png
Fuzzyball, fabric 03
deviation map
DETECTED, confidence 1.00
0078_025_03_x1024_y0.png
Warp ball, fabric 03
deviation map
DETECTED, confidence 1.00

defect-free patch, fabric 04

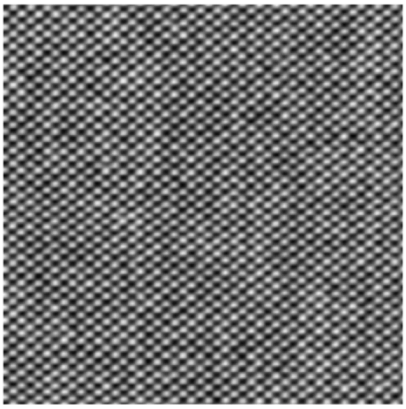


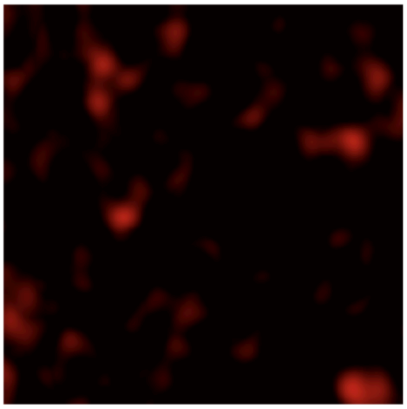


correctly clear, confidence 0.15

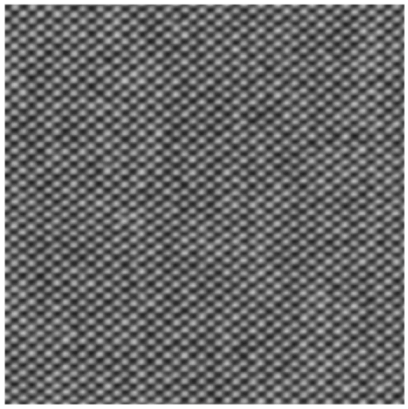


defect-free patch, fabric 03

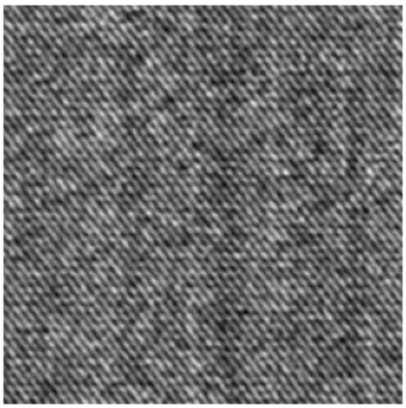


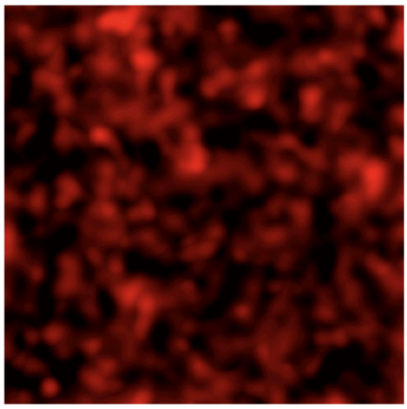


correctly clear, confidence 0.27

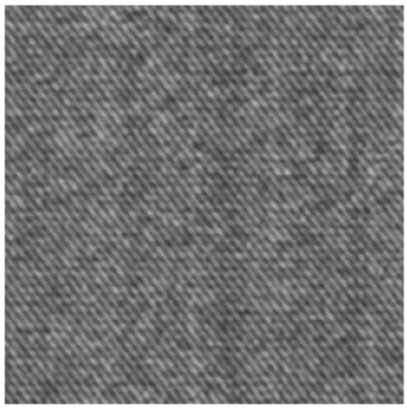


defect-free patch, fabric 05

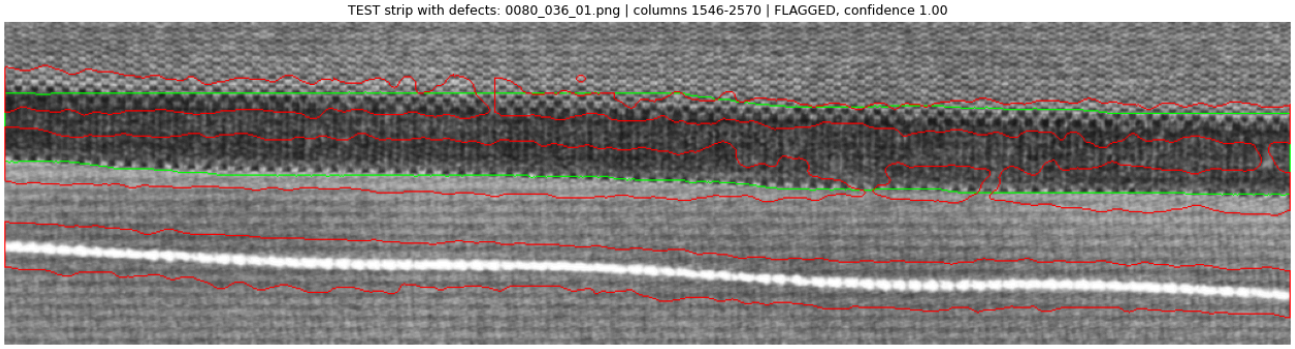




correctly clear, confidence 0.51



TEST strip with defects: 0080_036_01.png | columns 1546-2570 | FLAGGED, confidence 1.00



3. Challenges Faced

- Variations in weave period from one fabric to another: We discovered that the primary weave period was varied between fabrics. Due to this, it was not appropriate to adopt a single window size for all fabrics. This difference was compensated for by using the reference spectra for the various fabrics, combined with windows for the various periods in the adaptive method.

- Small defects: About 74% of true defects found were smaller than 32×32 window. The defects would only be very small so it was not easy to find and locate all the time.

- For the Week 9 edge analysis, the margin of false alarms on held-out defect-free patches (45%) was much smaller than the 16% margin that would be expected in random locations. From this observation, we chose a 16-pixel border margin. But this also implies that defects very close to the image edges will not be scored.

- Tuning resulted in an improvement in the DEF defect detection rate and image recall. The false alarm rate also rose however and at the same time the pooled pixel F1 decreased below the adaptive method that was not tuned. This indicates that there is no direct relationship between improved metrics and improved detection process.

- Equalisation, using CLAHE or not: We also compared the current detection results with the results obtained with blurring only to check if the equalisation was improving results. The chosen settings got an objective of 0.617 when using the CLAHE method, and 0.608 when not using the CLAHE method. No big difference, and current patch set is marginally better, so retained existing patch set rather than creating new patches.

- Combination of Gabor responses and Fourier method: We tested if a fusion of the two could help in detection. The Week 9 fusion results, however, were 0.671 (tuned Fourier) and 0.498 (maximum fusion) objective. We did not use Gabor fusion in the final approach because it was not as successful as the combined approach.

- Limitations of repeating texture: We will use a method that is based on determining and blocking out the repetition of a texture in a cloth. As a result, a defect which mainly alters the lightness of a weave without altering the overall structure of the weave can be difficult to be seen.

4. Literature Review

The AITEX Fabric Image Database is used for research on fabric-defect detection. It includes pictures of 7 normal and defective fabrics and defect masks [1]. Here, by using these masks, we can compare the defect regions that we predict to the actual location of the defects.

Wovens are useful for Fourier analysis since the normal texture is normally repetitive. Chan and Pang go on to say that defects can disrupt this regular pattern and produce changes in the frequency spectrum [2]. This is in relation with our Week 9 approach of finding the predominant weave frequencies and cancelling them out to reveal the unusual frequencies.

Campbell and Murtagh also talk about their use of Fourier features to detect defects in woven textile. Their approach consists of two steps: the first is a local defect detection, and the second is a contextual and morphological processing to

detect larger defects [3]. This is important to our work because we also produce residual maps and then threshold and clean up to find any possible defects.

In [4] Jing et al. investigated how Gabor filters could be used to detect fabric defects and LBP and Tamura features could be used to classify the texture of a fabric. In Week 9 they have two of the methods tested: Gabor and LBP, giving a reference to them. These texture-based methods were compared to allow us to see how well they perform on our dataset with Fourier-based methods.

Hanbay, Talu and Özgüven summarized the various methods used for detection of fabric defects such as structural, statistical, spectral, model-based, and machine learning.

d, learning-based and hybrid methods [5]. They also discuss factors such as accuracy, computational cost, reliability and sensitivity to noise. This is applicable to our project as we evaluate the methods with different metrics and also other concerns like false alarms and false defect detection.

References

[1] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, “AFID: A public fabric image database for defect detection,” AUTEX Research Journal, 2019. https://doi.org/10.2478/aut-2019-0035

[2] C.-H. Chan and G. K. H. Pang, “Fabric defect detection by Fourier analysis,” IEEE Transactions on Industry Applications, vol. 36, no. 5, pp. 1267–1276, 2000. https://doi.org/10.1109/28.871274

[3] J. G. Campbell and F. Murtagh, “Automatic visual inspection of woven textiles using a two-stage defect detector,” Optical Engineering, vol. 37, no. 9, pp. 2536–2542, 1998. https://doi.org/10.1117/1.601692

[4] J. Jing, H. Zhang, J. Wang, P. Li, and J. Jia, “Fabric defect detection using Gabor filters and defect classification based on LBP and Tamura method,” The Journal of The Textile Institute, vol. 104, no. 1, pp. 18–27, 2013. https://doi.org/10.1080/00405000.2012.692940

[5] K. Hanbay, M. F. Talu, and Ö. F. Özgüven, “Fabric defect detection systems and methods—A systematic literature review,” Optik, vol. 127, no. 24, pp. 11960–11973, 2016. https://doi.org/10.1016/j.ijleo.2016.09.110